

ActInf Textbook Group ~ Cohort 3 ~ Meeting 22 (Chapter 10, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 22 (Chapter 10, part 1) | 55:20

hi everyone it's cohort three

August 8th and we're in our first

discussion on chapter 10.

so before we look to the text or any of

the specific questions

um

would anyone like to just give any

comments or thoughts on Ted

hey Daniel I I'm pretty yeah I'm not

prepared for chapter 10 so it's better

I'm gonna actually means I I'll read uh

later

no sorry bye

well

chapter 10 has no

figures or equations it's it's a lot of

um synthesis and

overview

anyone want to give any comments or or

we can like look over it and try to pull

out some questions pull out some

thoughts

what do you think is the relevance of

this quotation in general we are least

aware of what our minds do best

is that true

I think we should first know what he

mean about the awareness

what is awareness

because the we just

suppose to this world but you know we

have so much different approach about

awareness about Consciousness about mind

some philosopher

they they don't accept this

assumption they think this is a we
should just work with neuron and bearing
Network neuron synops and no extra vote
because the
nobody can exactly know what is
awareness
we have different
introduction about awareness
why would Minsky say this
it reminds me of the um
the the discussion of uh the transparent
and opaque mental processes
with transparent mental processes are
like air like you see through them which
is to say you can't see them
opaque you can see them by virtue of
them blocking the something of light
so it's kind of like the fish in the
water
doesn't know about the water and so the
processes that are transparent to us
the things that are the most routine the
most mundane
we have this at least quote is saying
we're going to have the least awareness
of these
I'm not fully sure because I'm I believe
there are certain things we are unaware
of that our minds do well
but also I believe there are things our
minds do well that we are aware of
so I don't know if the general claim can
be said that we're least aware
of what we do best
since I think there's things that we're
aware of or not that we do well or not
but it's kind of a side point but but
it's you know as with all these quotes

they just start the chapter with kind of
a fun provocative
declaration
all right well
chapter 10 is a really interesting
chapter
they wrap up active inferences main
theoretical points that's part one
chapters one through five
and the Practical implementations
chapter six to ten
then they connect the dots abstract away
from the specific active inference
models discussed in previous chapters to
focus on integrative aspects of the
framework
all right
this is another
another spicy
peace
spicy and contentious
does active inference provide a complete
solution to the Adaptive problems that
sentient organisms have to solve
foreign
at all of anything
I mean we can vote and decide
but but but
what would be another way to say it or
how how would people yeah SML please
yeah you asked amazing questions
this approach I mean actively Affairs
can't solve everything but the
if nobody know about a future but at
least this optimistic approach is so
better than religious we just suppose
[Music]
everything is like a mystery

with this with this Assumption return we
can order so much uh thing for research
and scientists and Physicians can
research about that but if we we think
everything is Mystery just we give
forbid people for researching and uh in
the end we can't we can't achieve
anything

so my Approach is optimistic about
not in the moment

this will be a long procedure in and
future is open

great

yeah

treating things as a mystery or
somewhere on the mystery continue on
there with some amount of uncertainty
then it plus action so let's let's
figure it out and then this is kind of
like do we think that we're on the path
to achieving

a solution

to the Adaptive problems that sentient
organisms have to solve

or is this going to be a fruitless path

I think we we're

um through our actions we're on the
optimistic path

but I think this is a really

I mean it's a bold claim

and the narrow interpretation is did the
textbook provide a complete solution

but that's a little too narrow and I

think the broader question as Mel that
you point to is like as a long term

fields

will we have a scope

of addressing adaptive problems that

sentient organisms have to solve I even
think about it
outside of sentient organisms
just to the Adaptive problems that
things have to solve
where some kinds of things are adaptive
sentient organisms but that's not all
kinds of things
and then also
but so that's that's big
um however the second part of the claim
um is a little bit uh clearer and I
think more supported
active inference therefore offers a
unified perspective on problems like or
phenomena like
perception action selection attention
and emotional regulation
which are usually treated in isolation
and Psychology and Neuroscience
so I think that is a super strong claim
that
um can be pointed to when people what
does it mean that it's a unifying theory
of perception cognition and action or
what what does it mean that that the
first principle's approach
it's like well if you go to a
neuroscience conference
you're going to see totally different
ontology narratives tools documents
philosophies
on even just like the visual system
the models that are developed for visual
perception select action selection and
attention
those models may have no
interoperability at all

now that's not to say that they can't be
part of an integrated fabric
of a pluralistic scientific accounts
so it's not like it it doesn't matter if
they're not Unified
but this comes back to the needs and the
scruffies
do we want
coherence within and across systems of
Interest
or everybody has kind of like a level
with a tap out and just say no that's
like too much integration but at least
with these phenomena that they point to
here
the book
uh and the literature demonstrate that
these different phenomena
can be considered
in an integrated model
whereas approaches that don't start from
a first principle's position
in general do not have that ability
then they're going to connect to some
established Theory
and finally
extend into some broader topics
okay
wrapping up
only 190 pages
only only
95 page flips
away
95 little epistemic actions
uh
we're at the the end of the this little
chapter of the journey
the book offers a systematic account of

theoretical underpinnings and practical implementations of active inference

uh on these pages

they summarize the chapters this is one reason why I think it may be helpful to have chapter 10 be read at the beginning because they're going to summarize like really clearly what each chapter does so chapter one that's an overview

chapter 2 the low road that's the how Bayesian statistics and the generative modeling

chapter 3 is the high road the Y self-organization and persistence far from equilibrium modeled through the Markov blanket

chapter four

gives formal details on active inference focusing again on the generative model and this time exploring it a little bit more fully in the context of variational inference

chapter five

gives three connected case studies from mammalian neuroscience and talks about how these different neural systems

can be modeled with active inference and then how once you model them separately with active inference you can kind of stitch them together really nicely

chapter six was the recipe for Designing active inference models

a little bit of a Playbook or a guidebook

that also brings up a lot of different questions and considerations about scientific modeling

chapter seven and eight
were respectively about discrete and
continuous time generative models
chapter 9
was about active inference in the
setting of model-based data analysis
giving a little bit of information about
how you'd go about
doing a database analysis why you might
do that and and what that looks like and
so in that way it's kind of like chapter
six about building the generative model
but chapter nine is gonna
pull back another level and look at
what do you do with data
thank you

10.3 connecting the dots

great damn it

so

then it decades ago

many decades ago

lamented

that cognitive scientists devote too

much effort to modeling isolated

subsystems

or even isolated phenomena so not even

just like oh the visual tract

but like

visual perception in this one setting

whose boundaries are often arbitrary or

even worse disciplinary

he suggested to try instead modeling the

whole iguana a complete cognitive

creature perhaps a simple one and an

environmental Niche for it to cope with

esmail

what do you think about that or what did

it reference

bring up for you
then it's always
I mean he always is speaking about the
extra word
he first asked were you imagine for
example Consciousness after that you
make for yourself problem
philosophy zombie
I think all of you know what I mean
foreign philosophy zombie but uh he has
first
you should um you you should suggest we
have something name a Consciousness
after that we can speak uh philosophy
zombie doesn't have Consciousness and
everything like mind like awareness I
told you the first submitting
and we should just research about a
specific experience experimental
equation about mind about bearing and
even he denied a phenomenal and every
phenomenal uh concept
I think his idea is so close to
emergence
this thing you you
accounted like perception sensation
and so on all of them are a kind of
emergence I mean
the difference
compared with neuron and brain but
this is just bearing and neuron and not
exteriority and something V effects on
the world and Vault effect on us
something like this and we don't need
I'm angry with Dennis we don't need so
much extra
vote suppose and after that we will fall
in the problem

awesome yeah

pretty cool how a lot of those topics
like emergence and systems properties
which come from the Natural Sciences
then connect back to consciousness
and awareness like like we discussed a
little earlier

so benefits of active inference you know
next time you're at a party and you know
hanging out relaxing

one benefit of active inference is that
it offers a first principal account of
the ways in which organisms solve their
adaptive problems so even beyond the
theoretical Unity of diverse cognitive
phenomena like perception attention and
so on

a benefit of active inference is that it
gives a normative approach
so not just an is but actually an ought
and says this is how it should
be modeled

um I think it's it's it's easy to
be startled by a normative approach
because it can feel like impositional
or constraining
but I think that's part of our
interesting

Journey

is to understand what this normativity
is

and and

um what exactly is being what what ought
to be what way

and I'm sure there's many ways to say it
but but um

there's sort of some there's some some
narrow odds and some broader odds a

narrow ought would be like for solving a statistical problem with variational variational free energy is the best that you're going to do

so that's kind of that's like saying you know if you're doing a linear regression you ought to use the L2 sum of squares Norm

um but but

there's also kind of this broader potentially less formalized normative stance

which is that one

ought to

extend from that narrow ought about variational free energy as the imperative for statistical modeling and that you then

ought to model

specific cognitive processes perception action attention and emotion as well as their mechanistic manifestations

in a unified way

now someone else could say well I have a different normativity

I think that specific cognitive processes should be modeled in different and income measure it ways

I guess that could be someone's normativity or somebody could say

I've never really thought about that I don't I don't think I have a strong prior

let alone a normative stance on whether things like should or shouldn't be integrated

but but again it's just an interesting

um topic and and how many times does
normative come up in the book
you know 30 times
that's um
that's more times than you'll see
normative outside of like a
deontological ethics
book
and here's what they explained right in
the very beginning
normative means there's some standard
against which behavior can be scored
this is that narrower sense
free energy they're saying ought to be
the right quantity
for perceptual modeling
but then there's this secondary odds
which is a little bit like
and so you ought to do things in a
unified way
but that's a little bit less formal
but here's here's that normative level
it's Central assumptions all processes
minimize variational free energy so it
goes kind of this narrow piece
for a given statistical problem vfe is
the best estimator
and then there's this like slightly
broader normativity which is and you
ought to use vfv
for everything
to the extent that they say that
the problems that things face that
organisms or or other things face can be
described as inactive
they're they're related to action in the
world
so that's very compatible with embodied

inactive extended and cultured
Etc
affective neuroscience and pragmatism
and also
the preferences
are defined explicitly over
observations
this connects to perceptual control
theory
and also just
not historically
by tethering or modeling
to sensory data
that's kind of like
where the robber hits the road with not
overfitting
like
there's more that could be said there
but
by making sure that variational free
energy is a function of our beliefs and
the real incoming data
that's a a very strong check
on not overfitting not like proposing
some auxiliary variable
and then being super precise about the
auxiliary variable
but then the actual empirical data we
may be not modeling well
again more lists
of cognitive phenomena
pretty similar here they insert looks
like memory and intention
it's like
slowly mutating and expanding the list
what can organisms do they can change
their belief
perception learning attention

internal States

or they can change the world

action

change your mind change the world

two ways

to produce free energy

the third piece of the organism is the

sensory data but we keep our Thumb off

the scale

so the only two things that the organism

can do is change its mind and change the

world changing the world then changes

the external States

and that has indirect effect on sensory

data

they talk about memory

they talk about planning

planning and memory

can be understood as a

two sides of the same timeline

they have different in our just just in

our Arrow of time going forward world of

course memory is different than planning

because memory doesn't have the same

causal impact on our action selection

but yet our memories can influence what

we do

so it's not that they're without causal

impact but in both cases whether it's

reaching into the past with memory or

reaching into the future with planning

we can consider that a deep temporal

model

and deep temporal models so so that

that's how you synthesize memory with

planning and situational awareness in

the moment deep temporal models

now we're going to bring in another

level of the synthesis which is well
what is that deep temporal model about
maybe that deep temporal model
says
it has something to do with
interoception and emotion
so just there in a few sentences
that is a synthesis of past present and
future
sensory exteroceptive sensory
interceptive and affective
so that's that's what they mean by
what's enabled
and these kinds of relationships might
be pointed to
in
another field of Neuroscience or in
other fields but if it came from
Neuroscience it's unlikely to transfer
to spaceships and computers and all
these other systems
and whether it did or didn't transfer it
would be kind of descriptive and elusive
but it wouldn't have a formal basis like
we know that active inference does
okay again they're just hammering the
point home
we don't start by defining cognitive
functions
and try to puzzle piece them together
we start by what they characterize as
the complete solution
or maybe we could modify the adjective
and the noun like a comprehensive
approach or a unified account
but we provide a complete solution to
the problems that organisms have to
solve and then analyze those Dynamics

or fine-tune those Dynamics
to develop properties
that remind us of cognitive functions
so like what this reminds me of is
um sequential Dynamics
this came up multiple times
so sequential Dynamics have been modeled
with the lock of ultra model
it's just a property of certain
dynamical systems it's it's not an
apriori atomic cognitive function
but then that becomes deployed in a
cognitive behavioral modeling setting
to sequential actions like writing
so writing isn't a some core canonical
function
properties from dynamical systems we can
connect to their cognitive
concordances
and all these are maps not territories
so you know maybe there is some module
in some system but but
that's not necessarily even the best way
to go about modeling it
this connects back to the whole iguana
as well as to for example JF cloutier's
work with robotics
which mechanisms permit a living
organism to perceive the world remember
it and plan
so once you really think about what
something needs to do and be
to persist amidst uncertainty
the whole iguana
then even if you have a very coarse
grained picture of the whole iguana like
it's not atom level you should hear you
know it moves um out of the sun when

it's too hot into the sun when it's too
too cold
that is getting closer
to an understanding of the mechanisms
that permit it to live
rather than infinite molecular
description
those certainly our our mechanisms but
but they're not
helping us understand from a cognitive
perspective what's happening
okay with the normative perspective
we can reapproach
analogies and trade-offs
that have been identified in in
different systems
so exploration and exploitation
this paper
um is it's actually a really nice one
Hills at all 2015 I believe has some
representations of explore and exploit
in body
and in mind
so like come up with 10 words you know
you could go red orange yellow green
blue purple like that's kind of like
exploiting colors
or you can have like a Divergent
thinking
and similarly foraging
um you could be exploiting one resource
or
um exploring across resources
understanding that in the context of of
active inference which we explored a
little bit in the epistemic
rat in the maze
but it helps us rethink that explore

exploit and how to model and approach it
so beyond the theoretical Unity
we also get a bridge of principled means
of understanding corresponding neural
computations

it offers a process theory that connects
cognitive processing to expected
neuronal Dynamics

so you'll notice in the SPM package
a lot of the simulations just as part of
the default outputs they include like
dopaminergic signaling
and so that doesn't you know dopamine
does multiple things even in the
mammalian brain does multiple things in
different organisms brains so
um it's not saying that's the one and
only trace of dopamine

however

we can connect

molecular

mechanisms

to statistical properties like surprise
now they're going to explore
implications of active for psychological
functions as if we were sketching a
psychology textbook

so it's kind of like

within the body of this textbook there's
the germ

of another textbook

and they're going to highlight
similarities and differences

another reason why it may be relevant to
read chapter 10 early

10-4

predictive brains predictive Minds

predictive processing

we've talked about this a fair amount
but rather than a world view in which
sensory data come in
and are processed and then make their
way to their final internal
representation and then are used to
engage action a sort of feed forward
model of
perception and action
inactive inference inheriting from
we have an ongoing generative model
is generating predictions expectations
anticipations about sensory data
and about action
and then incoming sensory data
are able to be juxtaposed as they say to
validate their hypotheses predictive
coding
and there's some subtlety with what's
predictive coding what's predictive
processing and so on
and then at a slower time scale
you have the model updating essentially
its weights through learning
that imperative
is self-evidencing
the best generative model is the one
that's least surprised by incoming
sensory data
that's the maximum model evidence
and so that is how this epistemic and
pragmatic
um dual imperative is Unified through
action and active inference
they write the term predictive
processing is used in a broader and less
constrained sense compared to active
so again probably a lot of things you

could say like active inference is
predictive processing plus action
active inference is predictive
processing with action under the free
energy principle
but they're more alike than not
and predictive processing is the gateway
drug
I mean
lots more debates and and topics on that
perception
our perception
is our generative model
of course we want to be careful with the
map in the territory but I always return
to vision
we don't pay attention to our isocate
we don't notice our blind spot literally
we fill it in and we have high
resolution color vision all around the
eye
even though the retina doesn't support
so for me cicades blind spot and
peripheral color vision
give absolutely tangible evidence
that our visual perception
is generated and fine-tuned
not received
and likewise for other sensory
modalities
that concept of perception as
subconscious or subliminal inference
is ancients
helmholtz and before
Bayesian brain again another very
closely Allied topic with active
inference after all we're talking about
brains sometimes and we're talking about

Bayesian statistics

so how is active inference related to

Bayesian brain

Bayesian brain refers to a family of approaches that are not necessarily integrated and make different empirical predictions it's kind of a broad family

it's not like one specific model

because he could also refer to a little bit more of a realist flavor the brain

does Bayesian computation or a little bit more of an instrumentalist flavor we use Bayesian statistics to model the brain

but in either and and also it can apply at different levels of explanation

but broadly

active inference is a Bayesian brain

hypothesis but it's not just about brains and it's a process Theory not a hypothesis but yes it's Bayesian it could be about the brain

but if you do Bayesian brain plus action under the fep

I I believe you also compose active

but active inference is more integrated it's more normative

and it is better described as a process Theory that's falsifiable but but that is very closely aligned with a hypothesis

they return to predictive coding

which was discovered as a video compression

algorithm

and only brought into the neurosciences

most empirically in the late 90s

so it's only been 24 years

since
some of these things have even come onto
the scene
in active inference action processing is
analogous to perceptual processing
it's all part of the generative model
in chapter five
we explored how top-down descending set
points predictions
can then induce action
that's the equilibrium point hypothesis
there's another discussion of sensory
attenuation
there was a box on that earlier in the
book it's kind of like the opposite of
attention is attenuation you know
turning it up is like more attention
turning it down is like attenuating
and there was discussion around how
of our so attenuating our attention to
sensory input
creates like it opens up a space of the
possible
where then we can undertake an action
selection
and then lock in that precision and
that's exactly what happens with an eye
cicade that's why we don't perceive the
isocate
High attention fixed gaze
and then it's like
we push on the clutch or pull off the
gas whether you drive automatic or
manual
um move the eye it's not noticed it's
not attended to but of course it's still
getting photons
and then once it's fixed again we

redeploy the attention
and so that's why our phenomenology of
vision
is basically fixed fixed
without
um and maybe some pathologies maybe it's
there's nausea
if there's a failure of that kind of
attenuation it might be very
unsettling
in the isocare setting but in the bodily
motor setting this is where
um some people
um are challenged with beginning motion
so they can't step off the gas
so then there's sensory inputs like no
you're in a chair you should just stay
um or or inability to stop motion
idea motor these are kind of broad
conceptual theories that active
inference brings a lot of rigor to
but they basically relate to mind and
body
actions are selected on the basis of
their predicted consequences
cybernetics
tote
test operate test exit we've also
connected octave to UDA observe
orientiside Act
perceptual control theory we also have
requisite diversity and good regulator
theorem
um basically everything related to
active inference is going to be able to
handle
with this specific note that the
preferences are over perceptions

which kind of grounds higher levels of the system

optimal control theory so the inbound perceptual and learning side of active inference the cybernetic entity that's like signal processing and sense making the outgoing dependency on the mark on blanket is connected to control theory

here's some discussions about control theory

importantly optimal motor control is kind of aligned with reward learning in that there's a cost or a value function it uses these economic metaphors

cost value reward utility

whereas

we use all the apparatus that we've described earlier we don't need to invoke these economic metaphors or but we can account for utility in decision making

here's the economic theories of utility maximization and reinforcement learning we can account for those

with pragmatic value

dynamic programming there's really cool modeling work that's using dynamic programming in active inference when when all people have is pragmatic value utility reward

then

they have to propose these ad hoc strategies

to get non-pragmatic Behavior

or to escape local Optima

like burn-in novelty bonus intrinsic reward

the list goes on

active inference we just specify it up
front expected free energy epistemic
plus pragmatic value

rather than trying to find the most
rewarding or the highest utility path
we try to find the most likely path
so instead of climbing to the mountain
top of the mountain who knows if we're
at the top of the mountain

we're taking the path of least action
and that's what connects all of these
discussions to Bayesian mechanics
that's what makes it physics based among
other things

is that it's a variational principle of
least action

for cognitive systems

so just like a baseball with certain
starting conditions and so on has a
certain Parabola that's the path of
the Bayesian updating with a given prior
and attention and incoming data stream
is going to update a certain way
on the path of least action now that
that path of least action may also be
described by the most rewarding
trajectory or the highest utility
trajectory or whatever

but it turns out that it's a lot more
General and generic to describe it as
taking a path of least action which is
something that physical systems do
rather than this kind of like
aspirational

highest utility approach

that's connected to decision making
there's no utility or cost function

inactive inference
we select action plans based upon their
probability
of happening
not based upon their probability of
providing a reward
and there's parameters like the shaky
hand parameter or the temperature on
decision making if you turn the
temperature to absolute zero
then you're going to select the action
that you think is better even if it's
only a little bit better
whereas if you have the shaky hand
parameter very high high temperature
decision making then you blur
differences in the expected free energy
of policies and you basically select the
policies at um
reinforcement learning
policy is used differently
model based and model 3rl
and policy gradient methods
which are similar
procedurally in certain ways to active
however the objective function is
different
it's a reward function that's being
reinforced reinforced it's not a
generative model
surprise bounding free energy explicitly
and there are some differences more
planning
behavior-bounded rationality
this kind of connects to type 1 type 2
deliberative dual Theory
rational economon
free energy theory of bounded

rationality

valence emotion motivation

now they're kind of exploring

broader phenomena

homeostasis allostasis interoceptive

processing

pretty cool that active inference is

very flexible

to describe visual input or

pockets coming over the network

or interception

again needs and scruffies the neat wants

to have a unifying model that can

describe all those input output systems

or measurement control systems

the scruffy says no interception's

different than vision and that's

different than the computer packet

attention salience epistemic Dynamics

attention is covert action it's internal

action and it's modeled as such

overt action is embodied action

we can operationally Define attention as

the Precision associated with sensory

input if you're not paying attention

the sensory input doesn't matter in the

sense that it doesn't update your model

in the limit of paying extreme attention

whatever data point comes in you will

update your beliefs to be that

that's related to gain control in the

Bayesian setting like in audio

engineering

salience is expected Information Gain

salience is in terms of an action or

policy while attention is an attribute

of beliefs about sensory input

rule learning causal inference fast

generalization

great quote by Rumi

these are like half page to one page

long but there's you know there's

there's so much here and then active and

other fields open directions

sociocultural Dynamics we're exploring

that a lot in the ongoing course active

inference for the social sciences

machine learning and Robotics

huge topic a lot of work in this field

are those the only two

in summary

ending with with the Tolkien Lord of the

Rings quote

alluding to the golden ring on the cover

of the textbook

we are now at the end of our journey

their aim was to offer an introduction

to those interested in using methods

conceptually and formally

did they do it

however it is important to emphasize

active inference is not something that

can be learned purely in theory

we encourage anyone who has enjoyed this

book to think about pursuing it in

practice

whether or not you pursue a

computational practice

we hope you will reflect on it as you

engage in active inference and

day-to-day life

you might be looking

to see something you're not sure what it

it might be eating what you know you

like you might be trying something new

might be turning down the shower heat if

it's too hot might be
not turning down the shower Heat
ultimately we are confident that you
will continue to pursue active inference
in some form
that's the book
and then there's the appendices
any kind of closing thoughts
textual tap chapter no figures no
equations
it's all it's like a whole undergrad
degree kind of compressed
it covers a lot
I guess uh next week
we can uh
return to some of these
discussions bring up some of these notes
into the questions ask new questions
cool
any last comments
thank you all see you next time
thank you very much bye